Data augmentation via prompting for Speech and Language Disorders - Project Report

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1 Abstract

A primary challenge in clinical linguistics is the lack of naturalistic data for Speech and Language Disorders. This study explores whether Large Language Models (LLMs) can mitigate this issue by recognising patterns of morphosyntactic impairment to generate synthetic data. The project aims to address the data scarcity problem to enhance the classification and automated diagnosis of Language Disorders. Specifically, we use a subset of the ENNI Corpus comprising narrative transcriptions from children with Developmental Language Disorders. This specific condition was selected because it offers the most robust collection of publicly available data among language-disordered populations. The study employs in-context learning to generate new clinical stories, a technique in which the model is presented with examples of how to perform the requested task.

Since initial experiments show that standard prompting techniques produce data that either deviates from real distributions or lacks natural variability, we developed a refined method to attempt ecological validity. This two-stage strategy is called Sample-Wise Multi-Step Decomposition. In the first stage, the model is instructed to retell a narrative from the perspective of a typically developing child to establish a realistic linguistic foundation. In the second stage, the model should modify the baseline by (hopefully) attending to specific clinical markers of the disorder.

This methodology would enable the generation of an age-matched synthetic corpus of 2400 narrative samples. By using this synthetic data to augment limited clinical samples, the study aims to improve the performance of automated screening tools and support the early identification of children with language impairments. Moreover, if this method proves successful, clinicians and researchers working with other low-resourced populations could benefit from this data augmentation strategy.

2 Overview

Developmental Language Disorder (DLD) is a neurodevelopmental condition characterised by persistent difficulties in the acquisition and use of language. It is a complex phenomenon exhibiting high levels of variability in its clinical presentation. It is diagnosed when language impairments emerge during development without being acquired or associated with known biomedical conditions (e.g., brain injury or neurodegenerative diseases). Current clinical consensus accepts that the co-occurrence of other neurobiological conditions does not preclude a DLD diagnosis; thus, linguistic difficulties can often be present alongside cognitive weaknesses in areas such as procedural memory, motor control, and executive functions.

Between the ages of 3 and 5, children undergo intensive grammatical and morphological development. During this period, specific linguistic phenomena such as over-regularisation (e.g., "breaked" instead of "broke"), question and passive formation, and negation become observable. For English speakers, key biomarkers of DLD typically include difficulties with:

- **Pronoun usage:** e.g., *"Me want cookies," "Her is tired."*
- **3rd person singular morpheme -s:** e.g., *"He want cookies."*
- **Past tense morpheme -ed:** e.g., *"Yesterday I go to school."*
- **Plural morpheme -s:** e.g., *"They have two shovel."*
- **Wh-questions:** Difficulty producing or understanding structures like *"Who did the boy see?"*
- **Embedded clauses:** Difficulty with complex syntax, such as *"The kid who played with you is nice."*

Because children are in such an active state of development, research indicates there is no single "optimal" age for screening. High developmental variability makes it difficult to distinguish a temporary late-talker from a persistent disorder between ages three and five. While screening tools can identify at-risk toddlers as early as 15 months, their predictive accuracy often declines as some children catch up to their peers. Consequently, a formal diagnosis of DLD is generally considered most reliable from age four onwards. Based on these findings, the present study focuses on data from children aged four and five to reduce diagnostic instability.

3 Pilot Study: Methods

3.1 Materials and Analytical Tools

The ENNI Corpus The Edmonton Narrative Norms Instrument (ENNI) is a standardised clinical tool designed to collect narrative data from pediatric populations aged 4 to 9 [**?**]. Hosted on the `TalkBank` system within the `CHILDES`

database [**?**], the ENNI provides a robust normative framework for evaluating the language development of typically developing (TD) children alongside children with DLD. Narratives are elicited through a series of wordless picture sets of increasing complexity, which allow researchers to assess a range of linguistic levels, from basic story grammar to sophisticated morphosyntactic structures. In this study, the ENNI serves as the primary source of human reference data for benchmarking our LLM-generated synthetic transcripts. Each subject file contains 6 narratives elicited with two sets of pictures: stories with a Giraffe and an Elephant (A) and stories with a Rabbit and a Dog (B). The stories range from a simple two-character story to a complex four-character story. Some files contain an additional trial story, making the number of stories per group: 107 for DLD and 276 for TD.

CLAN and the KIDEVAL Utility Transcripts were analysed using the Computerised Language Analysis (`CLAN`) program, a specialised software suite developed for the automated processing of data in the `CHAT` (Codes for the Human Analysis of Transcripts) format. A critical component of our analytical pipeline is the `KIDEVAL` utility within `CLAN`. `KIDEVAL` is an automated clinical tool that compares a target transcript against age-matched normative data from reference databases (including the ENNI).

The utility provides a multidimensional profile of linguistic competence by calculating several key measures, including:

- **Syntactic Complexity:** Mean Length of Utterance (*MLU*) in morphemes and words, along with Developmental Sentence Scoring (*DSS*) and the Index of Productive Syntax (*IPSyn*).
- **Lexical Diversity:** Type-Token Ratio (*TTR*), Number of Different Words (*NDW*), and Vocabulary Diversity (*VOCD-D*) Optimum Average for measuring lexical richness.
- **Morphosyntactic Accuracy:** Frequency rates of grammatical markers such as present progressives, plurals, past tense, and auxiliary/copula verbs.
- **Disfluency and Error Markers:** Automated counts of retracings, repetitions, and specific morphological error tags.

By utilizing these standardised tools, we ensure that our synthetic data is evaluated using the same clinical metrics applied to human pediatric speech.

3.2 Experimental Setup

Model Configuration and Dataset The generation of synthetic clinical data was performed using the `Meta-Llama-3.2-8B-Instruct` model. To ensure reproducibility, we fixed the random seed at 42 and set the temperature to 0.7. The primary task involved the retelling of the ENNI B1 narrative task.

The synthetic corpus consists of 180 texts across six experimental conditions (30 texts per condition). For validation, we compared these against a human baseline comprising 62 children: 16 diagnosed with DLD and 46 Typically Developing (TD) controls, sourced from the ENNI and Gillam corpora.

Experimental Conditions We designed six prompting conditions to evaluate the transition from zero-shot logic to intensive in-context learning:

- **A0 (Instruction-based):** Zero-shot generation based on specific linguistic constraints.
- **B0 (Knowledge-based):** Zero-shot generation leveraging the model's latent clinical knowledge of DLD pathology.
- **C02–C10 (Few-shot):** In-context learning using 2, 4, 6, and 10 examples, respectively.

Example Selection: Stylometric Clustering Exemplars for the few-shot conditions were selected using a Stylometric Clustering strategy. We targeted transcripts from 4- and 5-year-old children from the ENNI and Gillam corpora. Examples were selected based on their proximity to centroids in a feature space defined by Mean Length of Utterance (MLU) and Type-Token Ratio (TTR). This ensured that the prompt exemplars were representative of the core linguistic profiles of DLD and diversified. For each few-shot prompt, we alternated examples between 4-year-old and 5-year-old subjects to provide a balanced developmental representation.

3.3 Evaluation

3.3.1 Distributional Similarity

To quantify the gap between the distribution of real clinical speech and our synthetic outputs, we employed the MAUVE metric [**?**]. MAUVE measures the divergence between the distributions of two text corpora by comparing their area under the divergence curve. To ensure robustness across different linguistic representations, we calculated scores using three distinct embedding models:

- **GPT-2:** A standard baseline for general-purpose language modelling.
- **RoBERTa-large:** For high-dimensional bidirectional context.
- **ChildBERT:** A domain-specific encoder pre-trained on stories for children.

Unfortunately, this analysis was inconclusive due to the very small number of samples, which pushed the model's score to premium optimism (invariant 0.99 in any sample and embedding combination). Therefore, we moved on to a different evaluation technique.

3.4 Conformal Anomaly Detection

3.4.1 Features Engineering

To represent the linguistic profiles of the subjects, we extracted a comprehensive set of morphosyntactic features (N=25). From `KIDEVAL`, we selected 22 morphological features. While `KIDEVAL` provides robust automated benchmarks for overall linguistic production, it is primarily designed to measure frequency and normative alignment rather than the specific nature of grammatical deviations. To capture the complementary side of the language impairment, we employed the standardised `CHAT` error-coding system to annotate the data. To address the sparsity of specific error codes in short narrative transcripts, we implemented a custom aggregation strategy for errors:

- **Agreement Errors (*agr_err*):** Aggregated markers for 3rd-person singular, uninflected, and variable subject-verb agreement.
- **Past Tense Errors (*past_err*):** Combined markers for omitted suffixes (*-ed*), over-regularizations, and irregular form errors.
- **Aspectual Errors (*ing_err*):** Omissions and misapplications of the present progressive suffix.

3.4.2 Conformal Anomaly Detection

Rather than performing a binary classification between clinical groups, we evaluated the "authenticity" of the synthetic data using Conformal Anomaly Detection (CAD). In the given context, CAD determines whether a synthetic sample is statistically indistinguishable from the real clinical population (given certain numerical features) or whether it constitutes a linguistic "outlier".

The detection pipeline was implemented as follows:

1. **Calibration:** An Isolation Forest was trained exclusively on the real clinical population (DLD, $N = 107$ stories) to model the reference distribution of obtainable anomaly score for genuine impaired speech. In particular a bootstrapping approach was followed to iteratively compute and eventually aggregate anomaly scores of different training realisations on out-of-bag instances. As an intermediate result, this yields a distribution of anomaly scores that is expected to represent anomaly scores for genuine impaired speech samples.
2. **Inference:** With the trained model and the reference (calibration) scores, the synthetic instances yielding valid p-values, computed as the relative rank of the obtained anomaly score of a synthetic instance within the previously generated reference distribution.
3. **Multiple Testing:** To provide statistical error control, the obtained batch of p-values for the synthetic instances were adjusted by the Benjamini-Hochberg (BH) procedure in order to control the proportion of false positives at a significance level $\alpha = 0.1$. Synthetic samples with an adjusted

p-value below this threshold were flagged as an anomaly and with that statistically distinct from the score distribution of genuine impaired speech (i.e. an unrealistic synthetic sample).

Feature Importance and Interpretability To understand which linguistic markers contributed most to a synthetic sample being flagged as an outlier, we performed Permutation Importance analysis. This involved calculating the median absolute deviation in outlier scores after shuffling individual feature columns across the bootstrap ensemble ($n = 100$ importance bootstraps). This allows us to identify whether specific LLM prompts result in "caricatured" or "atypical" impairments in categories such as past tense or subject-verb agreement.

4 Results

4.1 Linguistic Homogeneity and p-value Invariance

A consistent finding across all experimental conditions was the narrow range of adjusted p-values, indicating significant linguistic homogeneity. In the zero-shot knowledge-based condition, the difference between minimum and maximum p-values was only 0.008. This suggests that the model generates repetitive instances of impairment that occupy a confined statistical space rather than reflecting the diversity of natural speech.

While outputs from zero-shot and two-shot prompts remained statistically conformal to the real distribution by staying above the 0.1 threshold, they exhibited an instructional rigidity that failed to simulate the stochastic variability inherent in human language. Although these low-shot prompts successfully guided the model into the correct clinical territory, they did not capture the wide range of impairment profiles observed in actual developmental language disorders. By contrast, higher-shot conditions from four to ten examples resulted in a loss of clinical validity. In these instances, the model over-indexed on specific morphological errors to the point of being flagged as a statistical outlier. These distributional trends are summarised in Table 1 and Table 2.

5 Revised Prompting Strategy

5.1 Sample-Wise Multi-Step Decomposition

To mitigate the problem of data variance, we devise a Sample-Wise Multi-Step Decomposition strategy. The core of this strategy is to break the complex task of simulating a disorder into two manageable linguistic shifts, resulting in a two-stage pipeline:

1. **Task 1**: The model retells a target story as a typical child. This establishes an age-appropriate syntactic and lexical foundation.

Table 1: Conformal Anomaly Detection Results across Generation Conditions ($\alpha = 0.1$)

Condition	**Flagged Rate**	**Median p_{adj}**	**Top Features**
0-Shot (Instr.)	0/30 (0.0%)	0.256	*agr_err*, *u-cop*, **-PRESP*
0-Shot (Know.)	0/30 (0.0%)	0.260	*agr_err*, **-PRESP*, *u-cop*
2-Shot	0/30 (0.0%)	0.270	*ing_err*, *agr_err*, **-PRESP*
4-Shot	24/30 (80.0%)	0.087	*ing_err*, *agr_err*, **-PRESP*
6-Shot	27/30 (90.0%)	0.050	*ing_err*, *c-cop*, *agr_err*
10-Shot	28/30 (93.3%)	0.044	*ing_err*, *past_err*, *det:art*

Note: p-values adjusted via Benjamini-Hochberg procedure. A sample is flagged as an outlier when $p_{adj} \leq 0.1$, indicating it is statistically distinguishable from the real DLD population. Feature importances indicate which markers contributed most to the non-conformity score. **Legenda**: agr_err = agreement errors; *u-cop* = uncontractible copulas*; **-PRESP* = present participle; *ing_err* = errors related to the progressive morpheme (*-ing*); *c-cop* = contractible copulas; *past_err* = errors related to the realisation of past forms; *det:art* = number of articles.

Table 2: Quantile Distribution of Adjusted p-values across Conditions

Condition	**Min**	**p05**	**p25**	**p50 (Med)**	**p75**	**p95**	**Max**
0-Shot (Instr.)	0.256	0.256	0.256	0.256	0.256	0.256	0.264
0-Shot (Know.)	0.260	0.260	0.260	0.260	0.278	0.351	0.359
2-Shot	0.270	0.270	0.270	0.270	0.340	0.373	0.412
4-Shot	0.086	0.086	0.086	0.087	0.093	0.236	0.562
6-Shot	0.050	0.050	0.050	0.050	0.052	0.133	0.288
10-Shot	0.044	0.044	0.044	0.044	0.060	0.114	0.163

Note: p-values calculated using Isolation Forest with Jackknife-plus Bootstrap ($n = 100$). The shift from higher median values in zero-shot/2-shot to values below the $\alpha = 0.1$ threshold in higher few-shot counts illustrates the transition from statistically conformal to non-conformal (outlier) generations.

2. **Task 2**: The model receives the output from Task 1 and transforms it to simulate DLD.

Within this framework, we adopt **Diverse In-Context Learning**, a technique which uses a dynamic instead of a fixed set of examples. Moreover, following [**?**], we reduce the generation batches to 10 TD stories per set and 10 DLD stories per TD generated story (10 TD $*$ 10 $\rightarrow$ 100 DLD generations).

To do so, we apply a rigorous counterbalancing design across different types and age buckets (4- and 5-year-olds) to attempt broad ecological coverage and prevent data leakage between the examples and the target narratives. Specifically, we use the two narrative sets (A and B) from the Edmonton Narrative Norms Instrument (ENNI) interchangeably: when generating a synthetic version of an A story, the exemplars are drawn from the B set, and vice versa, as shown in Table 3.

Moreover, we implement a granular age-matching strategy for the generation

Table 3: Cross-Story Counterbalancing and Functional Exemplar Logic

Condition	**Target** (T) (Story)	**Exemplar 1** ($E1$) (Adult → TD)	**Exemplar 2** ($E2$) (TD → DLD)
Set A Targets	A1	B1	B3
(Using Set B)	A1	B2	B1
	A2	B1	B3
	A2	B3	B1
	A3	B2	B3
	A3	B3	B2
Set B Targets	B1	A1	A2
(Using Set A)	B1	A2	A1
	B2	A1	A3
	B2	A3	A1
	B3	A1	A2
	B3	A2	A1

of the synthetic corpus. Rather than employing broad age categories, we paired human exemplars from the TD and DLD groups by their precise chronological age (year;month.day), as detailed in Table 4. This strategy accounts for the rapid linguistic development occurring between the ages of 4 and 5, ensuring that the typical baseline and the clinical transformation provided in the prompt are developmentally aligned. We utilise a nested generation approach to maximise corpus diversity:

1. For each of the 24 unique child-pair configurations, we generate 10 distinct TD baselines.

2. For each of those TD baselines, we generate 10 distinct DLD variations.

This results in 100 synthetic DLD narratives per configuration. As summarised in Table 4, this results in a total corpus of 2,400 narratives. The generation pipeline is illustrate in Figure 1.

Table 4: Detailed Age-Matched Configurations and Cross-Story Mapping ($N = 2,400$)

Target Story	E1: Adult → TD ID	Age	Story	Gen.	E2: TD → DLD ID	Age	Story	Gen.
4-Year-Old Cohort (1,200 Stories total)								
A1	407	4;02.18	B1	10	444	4;02.04	B3	100
A2	445	4;04.09	B1	10	480	4;04.05	B3	100
A3	419	4;05.29	B3	10	478	4;06.04	B2	100
A1	411/451	4;06.20	B2	10	482	4;06.20	B1	100
A2	406/469	4;07.15	B2	10	427	4;07.16	B1	100
A3	459/415	4;08.04	B3	10	477	4;08.03	B2	100
B1	464/455	4;08.09	A1	10	475	4;08.12	A3	100
B2	405/461	4;08.15	A2	10	474	4;08.18	A1	100
B3	471	4;09.07	A2	10	476	4;09.07	A3	100
B1	472	4;09.07	A1	10	479	4;09.08	A2	100
B2	439	4;11.06	A3	10	413	4;10.26	A2	100
B3	443	4;11.15	A3	10	420	4;11.16	A1	100
5-Year-Old Cohort (1,200 Stories total)								
A1	528_2	5;00.24	B3	10	574	5;00.26	B1	100
A2	510	5;01.12	B3	10	573	5;01.05	B1	100
A3	545	5;01.12	B2	10	529	5;01.11	B3	100
A1	555	5;02.24	B1	10	567	5;01.12	B2	100
A2	552	5;04.05	B1	10	575	5;02.25	B2	100
A3	554	5;05.01	B2	10	531	5;04.07	B3	100
B1	511	5;05.10	A3	10	572	5;05.00	A1	100
B2	544	5;05.21	A1	10	535	5;05.09	A2	100
B3	517	5;05.25	A3	10	568_2	5;05.21	A2	100
B1	516	5;07.14	A2	10	576_2	5;07.11	A1	100
B2	532	5;07.29	A2	10	569	5;07.25	A3	100
B3	530	5;09.12	A1	10	570	5;09.21	A3	100
Totals	**24 Configurations**			**240**		**Grand Total:**		**2,400**

Note: The "Gen." columns indicate the number of synthetic outputs per configuration. For each TD exemplar, 10 synthetic baselines are generated; for each clinical exemplar, these are expanded to 100 DLD narratives (10 variations per baseline). Across 24 configurations, this yield totals 2,400 stories.

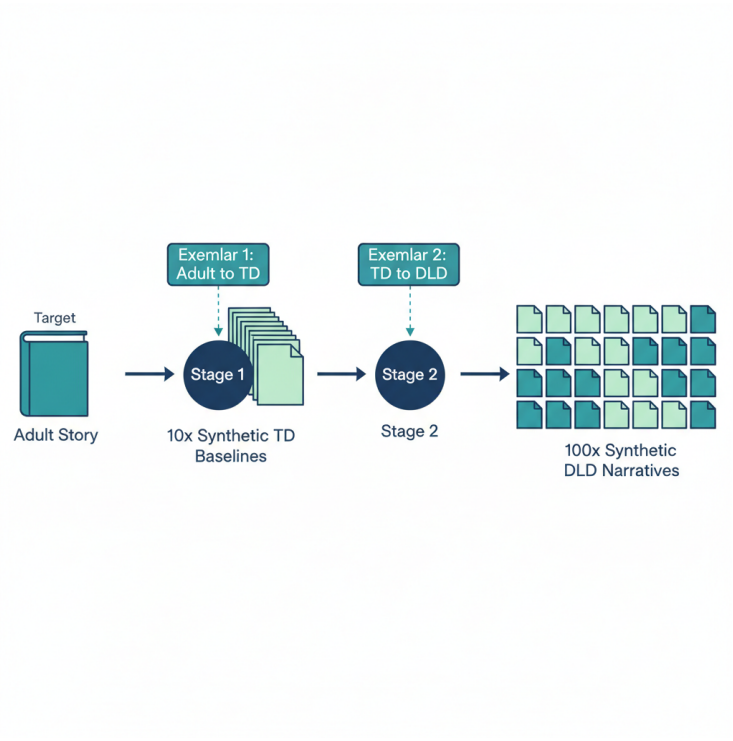

Figure 1: Sample-Wise Multi-Step Decomposition.